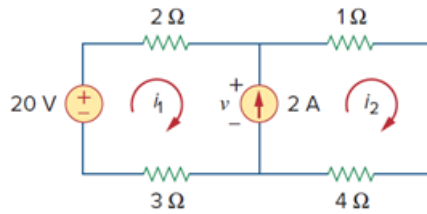


Problem

In the circuit of Fig. 3.49, current i_1 is:

- (a) 4 A
- (b) 3 A
- (c) 2 A
- (d) 1 A

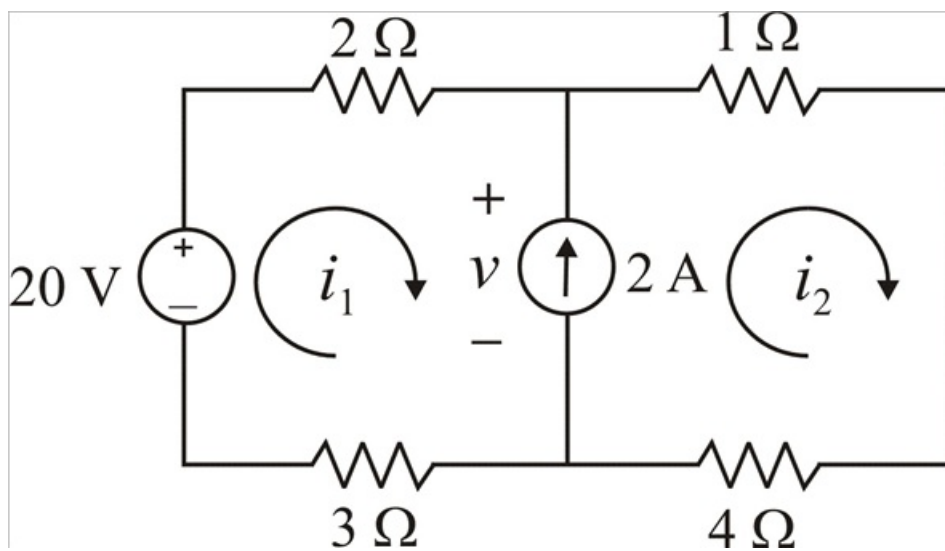
Figure 3.49:



Step-by-step solution

Step 1 of 5

Given circuit diagram is



Step 2 of 5

Above circuit is having two meshes.

current source is common for both the meshes, hence here supermesh forms.

$$\Rightarrow i_2 - i_1 = 2$$

$$i_1 - i_2 = -2 \dots\dots (1)$$

Step 3 of 5

By applying Kirchhoff's voltage law to outer loop we get

$$-20 + 2i_1 + 1i_2 + 4i_2 + 3i_1 = 0$$

$$5i_1 + 5i_2 = 20$$

$$i_1 + i_2 = 4 \dots\dots (2)$$

Step 4 of 5

By adding equations (1) and (2) yields

$$2i_1 = 2$$

$$i_1 = \frac{2}{2}$$

$$i_1 = 1 \text{ A}$$

Step 5 of 5

Given options are

(a) 4 A (b) 3 A (c) 2 A (d) 1 A

Therefore correct option is